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Strontium ion concentration effects on structural and spectral properties of $Li_4Sr(BO_3)_3$ glass

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ABSTRACT

Optimizing the concentration of host borate glass system to achieve a superior thermal and structural stability is challenging for sundry applications. A series of lithium strontium borate (LSBO) glasses with composition of $(85 - x)\ H_3BO_3 - 15Li_2CO_3 - xSrCO_3$, where $x = 5$, 7.5, 10, 12.5 and 15 mol% are prepared via melt quenching technique. Synthesized glasses are thoroughly characterized using XRD, FTIR, DTA, FESEM, PL, and UV–vis-NIR measurements to determine the influence of strontium (Sr^{2+}) ion concentration on thermal, physical, and structural properties of the glasses. XRD pattern confirms the amorphous nature of all samples. The FESEM images verify their homogeneous and transmitting surface morphology. Physical properties are determined in terms of glass density, molar volume, molar refractivity, polaron radius, inter-nuclear distance, field-strength, and ion concentration. Glass density is found to increase from 2.53 to 2.95 g/cm^3 with increasing Sr^{2+} ion contents. FTIR spectra exhibit the presence of two fundamental peaks in the range of 700–1070 cm^{-1} corresponding to the trigonal and tetrahedral stretching vibrations of BO_3 and BO_4 units. These peaks show a shift with the increase of modifier concentration. DTA results display peaks for glass transition, crystallization and melting at 500, 600 and 900 °C, respectively. Prepared samples are highly stable with Hurby parameter ~0.5. The direct, indirect band gap and Urbach energy calculated from the absorption edge of UV–vis-NIR spectra lie within 3.4–3.8 eV and 3.84–3.93 eV, 3.84–3.25 eV, respectively. The observed increase in refractive index from 2.17–2.19 is ascribed to the conversion of BO_4 into BO_3 units. Room temperature PL spectra under 430 nm excitations display two peaks centered at 482 and 526 nm accompanied by slight peak shift towards the lower wavelength due to the formation of new complexes in the glass network. Results are analyzed via different mechanism and compared. Excellent features of the results nominate these compositions potential for solid state lasers, photonic devices, and optical fibers applications.

1. Introduction

Several natural and synthetic borates are used in assorted industrial applications due to their qualified finishing products such as boric acid, anhydrous boric, anhydrous borax, borax pentahydrate, borax decorate, borax decahydrate, and sodium perborate in recrystallized units. However, understanding variability of borate crystal chemistry remains challenging [1]. Borate glass is promising due to various notable features including effective atomic number, high transparency, low melting point, and high-temperature stability [2,3]. Natural lithium borates are in the form of white powder with indistinctive order having melting point of 917 °C, solubility in the moderate range 1–10% and density of 2.4 g/cm^3. They are technologically beneficial as piezoelectric crystals [4]. Various alkaline and alkali earth metals are used as a second modifier to alter the glass structures by opening up modified network, to reduce the bond strength and to minimize the stickiness of the glass [5]. The high chemical durability, thermal stability and the most stable active ion hosts of oxide glasses make them favorable for optical and sensing applications [6,7]. The structural changes induced by Sr^{2+} incorporation are established.

Much work has been made on the borate glasses, but the work on glasses containing Sr^{2+} is found very few. According to the best of the authors' knowledge, for the first time the effect of Sr^{2+} ion concentration variation on the lithium-borate glass network structure and subsequent modification of their optical properties are systematically scrutinized. In this work, we employ SrO as a second modifier to enhance the release of electrons and to reduce the hygroscopic nature of borate glass [8–11]. Different composition of (LSBO) prepared to enhance the optical properties. The influence of Sr^{2+} ion concentrations on the thermal stability, physical and structural properties are determined via systematic characterizations. Results are explained using various mechanisms and a correlation between structural features and Sr^{2+} contents is established.

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2. Experimental

2.1. Samples preparation

The $Li_4Sr(BO_3)_3$ glasses of compositions $(85 - x)$ H_3BO_3 + 15% Li_2CO + x SrO_2 (x = 5, 7.5, 10, 12.5 and 15 mol%) were synthesized using melt-quenching technique at different concentrations of strontium ions. The powder of the compounds was weighted and well-mixed using milling machine. The mixture was melted in an alumina crucible for 1 h using an electric furnace at a temperature of 1300 °C. After completion of melting, the liquid glass was poured and quenched on well-polished pre-heated steel plate. Then, the samples were annealed at 400 °C for 3 h to eliminate the mechanical stress. Five samples (S1, S2, S3, S4, and S5) with their compositions are summarized in Table 1.

2.2. Samples characterizations

The amorphous nature of all samples (at room temperature) was verified using X-ray diffraction (XRD) analysis (Siemens Diffractometer D5000), equipped with diffraction software analysis. It used Cu Kα radiation (λ = 1.54 A) operating at 40 kV and 30 mA. The XRD profiles on powdered samples were collected in the range of 2θ = 5–90° at scanning rate of 0.05°/s. The particle morphology, purity, and the phase homogeneity of these glasses were analyzed via field emission scanning electron microscopy (FESEM). Infrared spectra for these glasses were recorded using a Scimitar FTS 2000 instrument (FTIR). FTIR measurements in the wave number range of 4000–400 cm^{-1} with instrument resolution of 0.8 cm^{-1} were carried out on thin pellet samples by grinding them with potassium bromide (KBr) of 100:1 mg ratio. The mixture was then pressed at 120 MPa pressure to obtain a transparent pellet with an approximate thickness of 2.0 mm and diameter 10.0 mm. The differential thermal analysis (DTA) was performed using Perkin Elmer Pyris Diamond Analyzer fitted with a copper target and nickel filter. The glass transition temperature (T_g) was measured using T_A instruments DTA 2010 differential thermal analysis with a programmed heating rate of 10 °C/min in the temperature range of 50–1000 °C.

The room temperature absorption spectra for all samples in the range of 200–2000 nm were recorded using a UV–vis-NIR Shimadzu 3101 Spectrophotometer. Physical parameters such as density (ρ), molar volume (V_m), refractive index (n), molar refractivity (R_m), polaron radius (r_p), inter-nuclear distance (r_i), field strength (F), ion concentration (N), and Urbach energy were determined from appropriate relations. Photoluminescence (PL) measurement was performed using a Perkin Elmer LS55 Luminescence Spectrophotometer, where xenon arc lamp of wavelength 300–1300 nm acted as excitation source. The luminescence signal was recorded via a Monk–Gillieson monochromator and a photo-diode detector.

3. Results

3.1. XRD pattern

Fig. 1 displays the XRD patterns of all prepared LSBO samples with different concentrations of SrO annealed at 400 °C. Complete absence of any sharp peaks verifies the amorphous structure of synthesized samples [12,13]. Furthermore, the occurrence of two peaks in the XRD pattern at around 20–30° and 40–50° are attributed to $LiBO_3$ and $SrBO_3$, respectively.

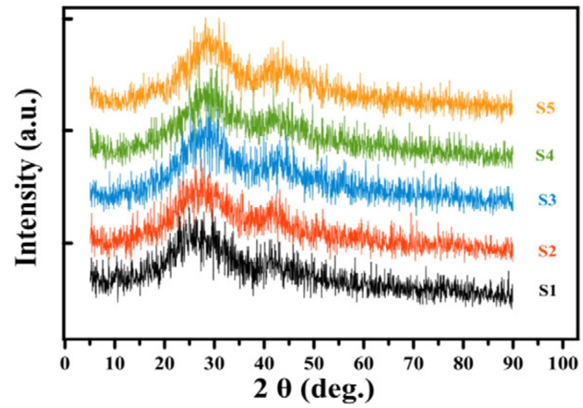

Fig. 1. XRD patterns of LSBO at different $SrCO_3$ concentrations.

Table 1
Compositions and codes of prepared glass samples.

Samples	Concentration (mol %)		
	Li_2CO_3	$SrCO_3$	H_3BO_3
S1	15	5	80
S2	15	7.5	77.5
S3	15	10	75
S4	15	12.5	72.5
S5	15	15	70

3.2. FESEM spectra

The FESEM images as illustrated in Fig. 2 (S1–S5) clearly display the absence of any grains and samples homogeneous morphology. An alteration in the glass network structure is clearly evidenced with the increase of SrO contents. This structural modification is noticeably reflected in FTIR spectra as well as in FESEM images.

3.3. FTIR analyses

Fig. 3 shows the IR spectra of all samples. The appearance of two fundamental peaks in the range of 700–1070 cm^{-1} is attributed to the trigonal and tetrahedral stretching vibrations. In addition, these peaks are shifted towards lower frequency with increasing the modifier Sr^{2+} concentration due to its high atomic weight than other atoms in the composite. Three major absorption bands are observed [14–16]. The modes at 700.14, 702.02, 728.31, 717.04 and 732.06 cm^{-1} are assigned to B–O–B bending vibration of stretching of trigonal structure of BO_3 and BO_4 groups [17,18]. The intensity of this band decreases with the addition of SrO. The band observed within 879 to 899 cm^{-1}, and 1040 to 1061.05 cm^{-1} are allocated to B–O bond stretching mode of BO_4 groups [19–21]. As the tetrahedral BO_4 groups are strongly bonded than the triangular BO_3 groups, therefore a compact structure is expected leading to a higher density [22]. The appearance of strong band around 1200–1500 cm^{-1} arouse from B–O symmetric stretching of BO_3 groups of ortho-borate units [23,24]. Intensity of all bands is decreased with the increase of SrO concentration. The occurrence of broad bands in the region of 3400–4000 cm^{-1} are assigned to the symmetric stretching of O–H groups (H–O–H) [25,26]. Table 2 enlists the FTIR band assignments. The source of error associated to the measurement are mostly instrumental and within the experimental range ~5%.

3.4. DTA thermogram

DTA measurements are performed to capture the information related to glass transition, crystallization, melting and sublimation temperature. Material under study together with inert reference material is subjected to same thermal condition, and the changes between two materials are recorded. Glass forming ability is estimated using following Kauzmann relation:

$$T_{rg} = \frac{T_g}{T_m} \tag{1}$$

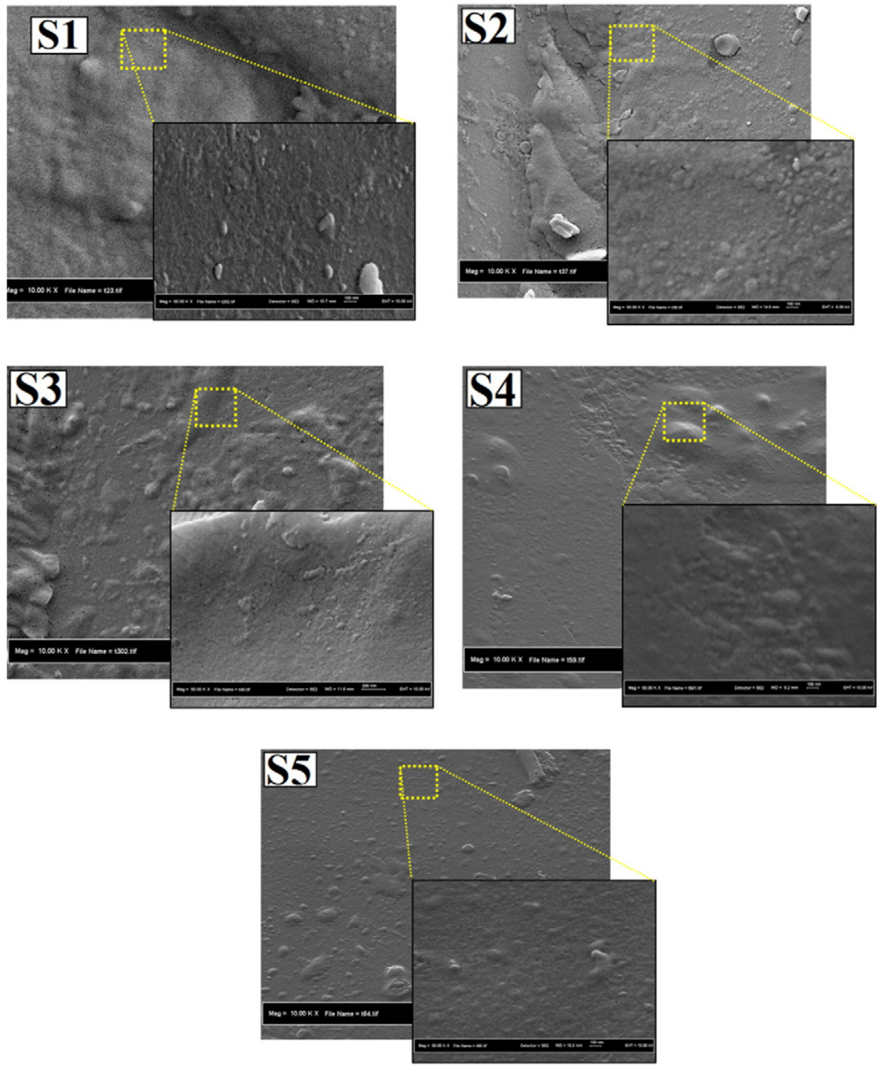

Fig. 2. FE-SEM images of LSBO with different ion concentrations of SrO for samples S1, S2, S3, S4, and S5.

where is the glass transition temperature and is the glass melting temperature. For good glass forming ability one must achieve $0.5 \le T_{rg} \le 0.66$. Present glass composition obeys the Kauzmann assumption with excellent values of T_{rg}.

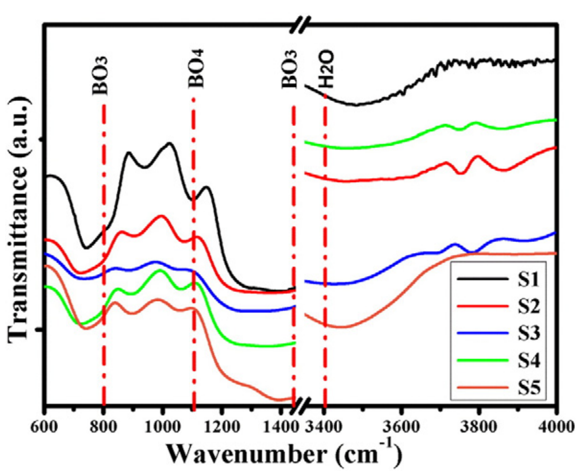

Fig. 3. FTIR spectra of lithium strontium borate glasses with different ion concentration of SrO.

Conversely, according to Hruby assumption, the glass thermal stability can be determined from,

$$H_g = \frac{T_c - T_g}{T_m - T_c} \tag{2}$$

where T_c is the glass crystallization temperature. For $H_g \le 0.1$, glass stability is very poor but for $H_g \sim 0.5$, it is superior [2]. The glass forming ability and stability for all samples are summarized in Table 3.

Fig. 4 illustrates DTA traces of all prepared samples exhibiting endothermic peak [13] corresponding to T_g values at 518.01 °C, 507.66 °C, 511.75 °C, 513.91 °C, and 511.99 °C. The exothermic peak for all five samples corresponding to T_c is revealed at 688.56 °C, 657.29 °C, 658.49 °C, 657.29 °C, and 666.19 °C. In addition, the endothermic peak corresponding to T_m is located at 889.67 °C, 888.71 °C, 894.01 °C, 897.13 °C, and 898.33 °C. The values of T_{rg} and H_g for samples S1, S2, S3, S4, and S5 are determined to be 0.58, 0.57, 0.57, 0.57, and 0.57 and 0.85, 0.65, 0.62, 0.59, and 0.66, respectively.

3.5. UV–vis spectra

Fig. 5 depicts the absorption spectra of all samples with varying SrO contents. Interestingly, the optical absorption edge is found to be shifted towards lower wavelength (380 to 352 nm) with the increase of strontium concentration. It is important to note that, the study of optical

Table 2
FTIR peaks positions of the lithium strontium borate glasses system.

Samples	Bond Assignment (cm^{-1})			
	B–O (stretching of trigonal BO_3 bond) ± 0.07	B–O (stretching of tetrahedral BO_4 bond) ± 0.04	O–H (H_2O) bond ± 0.1	Bending vibration ± 0.03
S1	1365.50	898.56, 1056.29	3445.59, 3739.18	700.14
S2	1373.01	894.80, 1060.05	3453.34, 3738.55, 3853.01	702.02
S3	1312.92	883.54, 1041.27	3443.34, 3701.65, 3781.71	728.31
S4	1346.72	891.05, 1052.54	3460.85, 3742.31, 3849.26	717.04
S5	1380.52	879.78, 1045.02	3427.70, 3861.14	732.06

absorption and band edge is useful to acquire information about the band structure as well as energy gap of crystalline and amorphous materials. This phenomenon involves the absorption of photon with energy higher than the energy band gap of the material. Thus, the absorption spectra provide the signature of atomic vibration and electronic properties of the material [28]. Following Mott and Davis [29], the energy gap expression can be written as,

$$E_g = \frac{hc}{\lambda} \tag{3}$$

where E_g is energy gap, h is Plank's constant, c is speed of light, and λ is the wavelength of incident photon. The photon energy ($h\nu$) dependent absorption coefficient (α) yields [28],

$$\alpha = (h\nu - E_g)^r \; \frac{A}{h\nu} \tag{4}$$

where r is constant depending on state, direct or indirect for direct 2 and for indirect 1/2 allowed transition, and A is a constant called band tailing parameter [30]. The excitation coefficient k is calculated using the expression [31],

$$R = \frac{(1-n^2)+k^2}{(1-n^2)+k^2}. \tag{5}$$

Urbach energy is estimated from the empirical relation,

$$\alpha(\upsilon) = \beta \exp\left(\frac{h\upsilon}{\Delta \mathrm{E}}\right)$$

where ΔE is the Urbach energy, β is obtained from the inverse slope of ln α versus $h\upsilon$ curve. Urbach energy being a measure of the width of localized states describes the degree of disorder in amorphous and crystalline system.

Table 4 summarizes the strontium ion concentration dependent energy band gap of various glasses. The value of energy band gap is increased from 3.85 to 3.87 eV with the increase of SrO concentration. Table 4 shows the variation of the band gap energy E_g and the Urbach energy Ln (α). A decrease in the optical gap energy is observed with the increase of SrO concentrations from 5 to 15 mol%. This is attributed to the difference in ionic radius of O, B and Sr. The direct band gap energy appeared wider due to the decrease in the transition tail width. This observation and the associated red shift effect is in agreement with other finding [32]. Moreover, the optical band gap energy and transparency are found to influence strongly by the SrO concentration changes.

Table 3
Strontium concentration dependent thermal properties.

Samples code	T_g (°C) ±0.01	T_c (°C) ± 0.01	T_m (°C) ±0.01	T_{rg} ± 0.001	H_g ± 0.001
S1	518.01	688.56	889.67	0.58	0.85
S2	507.66	657.29	888.71	0.57	0.65
S3	511.75	658.49	894.01	0.57	0.62
S4	513.91	657.29	897.13	0.57	0.59
S5	511.99	666.19	898.33	0.57	0.66

The decrease in the Urbach energy is attributed to the creation of less number of defects as reported [33,34].

The optical band gap energy increases with increase in the SrO concentration. This increase is due to the structural changes undergoing in studied glasses. Initially, at 5–15 mol% concentration, the increase in band gap energy attribute to large number of tetrahedral BO_4 unit conversion to trigonal BO_3 units of borate. By adding SrO to glass systems, excess oxygen ions will available to utilize to convert BO_3 to BO_4 groups. In glasses network, the bond of tetrahedral BO_4 groups is stronger than that of BO_3, which is responsible for the increase in optical band gap [35].

The values of E_g obtained by extrapolating $(\alpha h\upsilon)^{1/2} = 0$ for indirect ($E°_{\text{indirect}}$) and $(\alpha h\upsilon)^2 = 0$ for direct ($E°_{\text{direct}}$) transition. Fig. 6 displays the tauc plot [36] between $(\alpha E)^2$ and photon energy ($h\nu$). This is generated to calculate the direct band gap energy $E°_{\text{direct}}$ (eV). The value of indirect energy band gap $E°_{\text{indirect}}$ (eV) is increased from 2.75 to 3.23 eV as the SrO concentration is increased. Fig. 7 displays the modifier concentration dependent variation of direct and indirect energy band gap.

The absorption coefficient shows an overall increase with the increase of SrO concentrations as shown in Fig. 8. This enhanced absorption is due to the activation of matrix with bigger sized strontium ions.

Fig. 9 illustrates the SrO concentrations dependent variation in the Urbach energy ranging from 3.84–3.25 eV (Table 4). The observed decrease in Urbach energy with the increase of SrO concentration is ascribed to the generation of less bonding defects and non-bridging oxygen [37]. The glasses with a smaller Urbach energy would have lower disorder and more compactness.

3.6. Photoluminescence spectra

Fig. 10 shows the room-temperature emission spectra of $Li_4Sr(BO_3)_3$ glasses at different concentrations of SrO. The main peak of the emission band for sample S1 is appeared at around 482 nm. The intensity and the FWHM of this band are gradually changed with the increase of SrO contents (Table 5). Furthermore, this peak displays a sudden shift towards a higher wavelength with intensity enhancement as the concentration of SrO increases to 12.5 mol% (S4). A quenching in the emission intensity

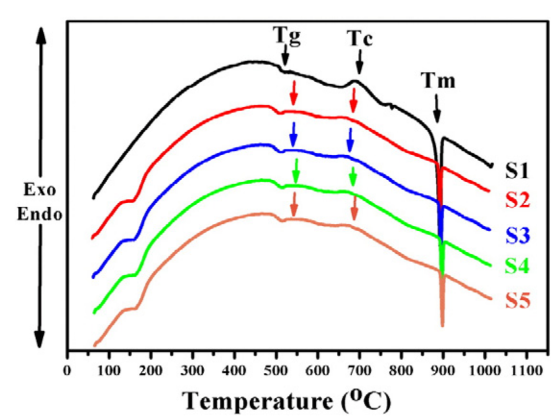

Fig. 4. DTA traces of all glass samples.

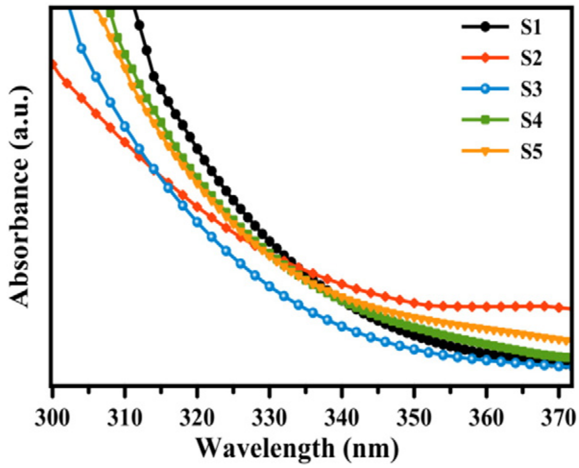

Fig. 5. Optical absorption of lithium strontium borate glasses with different ion concentrations of SrO.

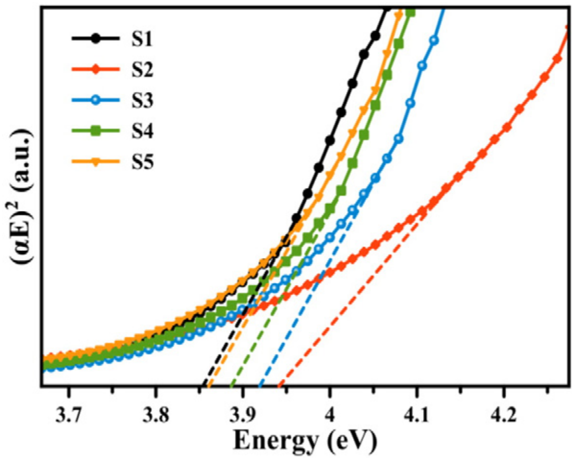

Fig. 6. Tauc plot for direct energy band gap determination.

accompanied by the peak shift towards a lower wavelength beyond this concentration is observed. This peak shift is attributed to the creation of new complexes in the glass with the addition of SrO [38].

Fig. 11 clearly shows that initially the PL intensity is quenched with increasing SrO concentration [39]. Then, it increased and again decreased. Sample S4 (12.5% concentration) revealed the maximum PL intensity. This decrease is majorly attributed to the SrO concentration quenching effects and the instability of the excited ions to the most upper level. These unstable ions are radiatively relaxed to the lower energy level and caused a change in the observed emission intensity.

3.7. Physical properties

SrO concentration dependent physical properties of all the prepared glasses are enlisted in Table 6.

Glass density (ρ) is determined using Archimedes' principle with toluene as immersion liquid (99.99% purity) from the expression,

$$\rho = \frac{W_a}{W_a - W_i} \times 0.896\ g\ cm^{-1} \tag{7}$$

where W_a and W_i are the weight of glass in air and liquid, respectively. The density of toluene at room temperature (27 °C) is 0.8696 g cm^{-3}.

Molar volume, Vm (cm^3 mol^{-1}) of glasses is calculated from,

$$V_m = \frac{M}{\rho} \tag{8}$$

where M is the molecular weight.

Fig. 12 illustrates the SrO concentration dependent variation in glass density and refractive index. The density is found to increase continuously from 2.53 to 2.95 g cm^{-1} with the increase of modifier SrO concentration. This increase is attributed to the following factors: (i) conversion of [BO_3] triangles into [BO_4] tetrahedral, (ii) enhancement of molecular mass of the glass due to insertion of higher atomic weight of strontium, and (iii) increase of the oxygen–boron ratio because of the increase in SrO concentration. Another indication of the enhanced glass structure compactness is the increase in density and simultaneous decrease in molar volume (32.6078–28.9184 cm^3/mol). This decrease is ascribed to the increase in inter-atomic spacing [33, 43]. The graph showing the increase in density with decrease in molar volume reveals a change in the structure of glass with increasing SrO contents.

The refractive index (n), molar refraction, and molar polarizability of the glasses are evaluated using [44–46],

$$\frac{n^2-1}{n^2+2} = 1 - \sqrt{\frac{Eg}{20}} \tag{9}$$

where Eg is the direct optical band gap obtained from UV–vis absorption edge.

Refractive index is found to decrease gradually with the increase of SrO concentration (Fig. 12). This decrease is attributed to the creation of less non-bridging oxygen atoms due to breakage of glass network structures [47]. A sudden decrease in the refractive index of 5 mol% of SrO may be due to the radical change in polarizability.

The molar refractivity (Rm) is calculated from Lorentz–Lorenz formula,

$$R_m = \left[(n^2-1) \div (n^2+2)\right] \times V_m \tag{10}$$

Table 4
Direct E°_{direct} (eV), indirect $E^{\circ}_{indirect}$ (eV) energy band gap, Urbach energy, absorption, and extinction coefficient for all samples.

Samples	E°_{direct} (eV) ± 0.001	$E^{\circ}_{indirect}$ (eV) ± 0.001	Absorption coefficient α	Extinction coefficient $K \pm 0.001$	Urbach energy $Ln\ (\alpha)$
S1	3.840	3.400	259,082.6	0.713	3.84
S2	3.940	3.200	338,047.7	0.968	2.23
S3	3.930	3.440	266,356.1	0.727	3.61
S4	3.890	3.350	340,603.2	0.922	3.33
S5	3.870	3.300	381,781.4	1.039	3.25

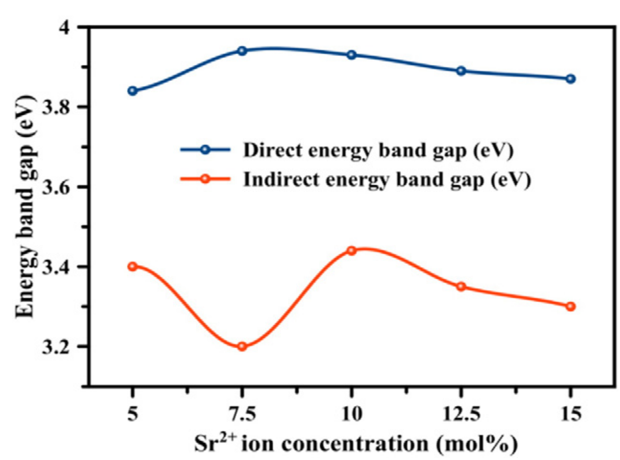

Fig. 7. Strontium ion concentration dependent of direct and indirect energy band gap variation.

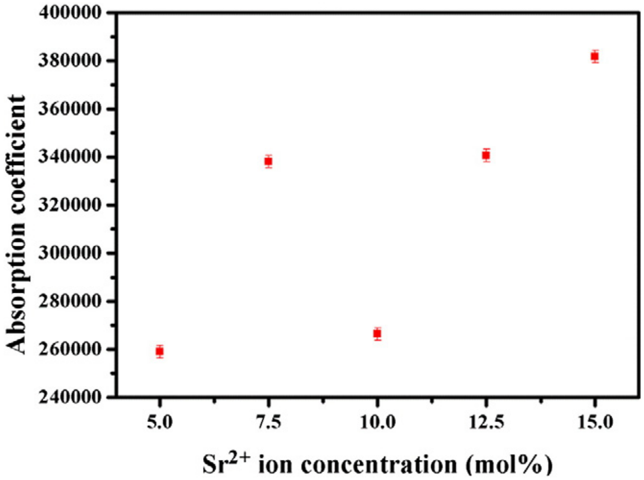

Fig. 8. $SrCO_3$ concentrations dependent absorption coefficient.

where the quantity $(\frac{n^2-1}{n^2+2})$ is called the reflection loss [48]. The molar refractivity is related to the glass structure, and it decreases monotonically with the increase of SrO concentration as shown in Fig. 13.

Following [49,50], the polaron radius is written as,

$$r_p\ (\AA) = \frac{1}{2}\left(\frac{\pi}{6N}\right)^{\frac{1}{3}}. \tag{11}$$

Inter-nuclear distance is given by,

$$r_i\ (\AA) = \left(\frac{1}{N}\right)^{\frac{1}{3}}. \tag{12}$$

The field strength is expressed as,

$$F = \left(\frac{Z}{{r_p}^2}\right). \tag{13}$$

The ion concentration is calculated using,

$$N = \frac{\text{mol}\% \times \rho \times NA}{Mi}\left(\frac{Ions}{\text{cm}^3}\right) \tag{14}$$

where mol% is mole percent of SrO, ρ is the glass density, NA is Avogadro number and Mi is the average molecular weight of the prepared glass [51]. The increase in field strength from (430–639 cm^{-1}) as shown in Fig. 14 is originated from the occurrence of strong link between Sr^{2+}

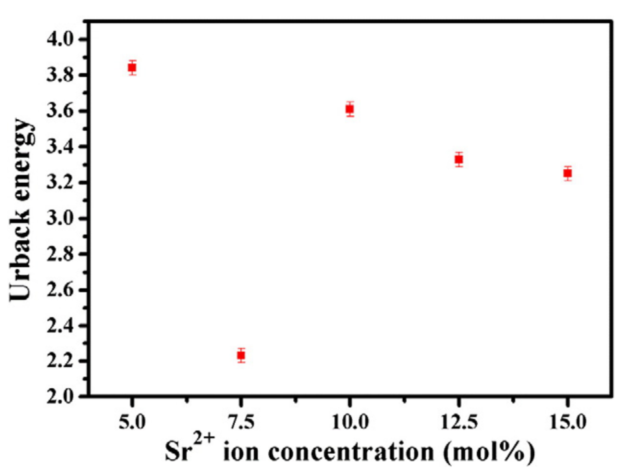

Fig. 9. SrO concentrations dependent Urbach energy.

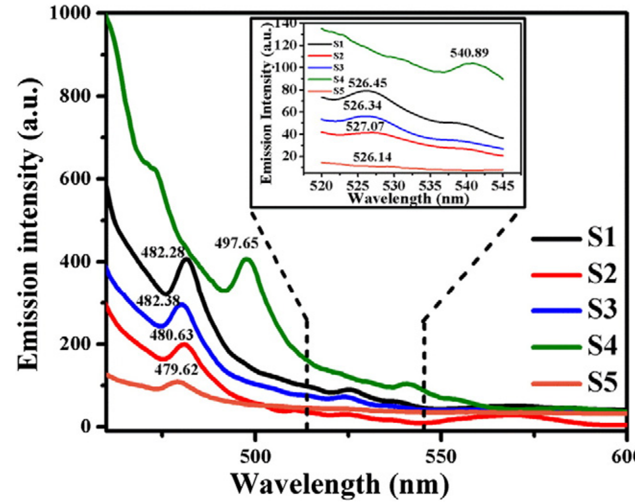

Fig. 10. Emission spectra of $Li_4Sr(BO_3)_3$ at different SrO contents.

Table 5
SrO contents dependent fluorescence intensity and FWHM of $L_4Sr\ (BO_3)_3$ phosphors.

Sr Contents (mol%)	λ	Intensity (a.u.)	FWHM
S1	482.28	398.01 ± 4.01	5.98 ± 0.03
S2	480.63	213.53 ± 2.13	5.12 ± 0.03
S3	482.38	285.68 ± 2.86	6.72 ± 0.03
S4	497.65	406.60 ± 4.07	6.99 ± 0.03
S5	479.62	110.92 ± 1.11	8.12 ± 0.04

and B-ions. This link is resulted from the possible displacement between Sr^{2+} and newly generated oxygen atoms from the conversion of BO_3–BO_4 units [52].

4. Discussion

Complete absences of any sharp diffraction peaks in the XRD pattern indicated that the prepared samples are truly amorphous in nature. It clearly indicates the absence of long range atomic arrangement and the lack of periodicity of three dimensional networks, where no well defined planes in the structure on or around which the constituent's atoms are regularly arranged. The homogeneous morphology and nonexistence of any faceted grain appearances in the FESEM image are in agreement with the XRD result. The transparent nature of the samples is

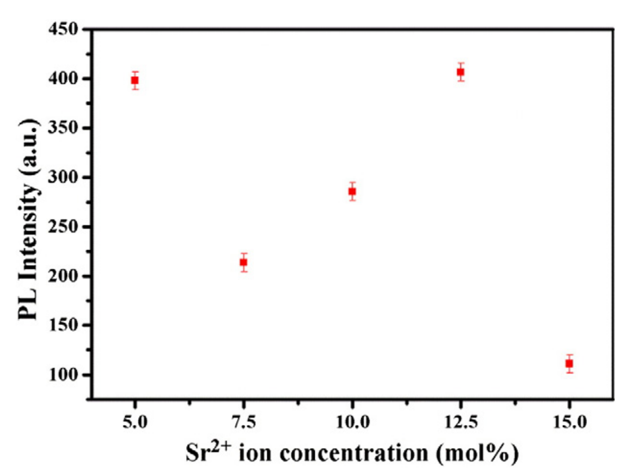

Fig. 11. SrO concentration dependent photoluminescence intensity of $Li_4Sr(BO_3)_3$ glass systems.

Table 6
The physical parameters of all prepared glasses [40,41]. The errors [42] in the measurement of density are estimated to be ±0.002 mm, 0.008 g cm^{-3}, for molar value and molar refractivity, reflective is ±0.004, and field strength is ±3.4.

Sr^{2+} (mol%)	ρ (g cm^{-3}) ±0.002	Vm (cm^3 mol^{-1}) ±0.01	n	Rm ± 0.01	MW	N (Ions/cm^3)	r_i (A°) × 10^{-8}	r_p (A°) × 10^{-9}	R ± 0.004	F (cm^{-1}) ± 3.4
S1	2.530	31.60	2.19	17.65	78.589	9.5277×10^{22}	0.1094	8.8219	8.066	430.0
S2	2.660	30.55	2.17	16.89	79.959	1.4781×10^{23}	9.4566	7.6206	7.872	498.0
S3	2.760	29.93	2.18	16.64	81.279	2.0117×10^{23}	8.5532	6.8766	7.969	552.0
S4	2.830	29.67	2.18	16.49	82.62	2.5371×10^{23}	7.8981	6.3647	7.969	597.0
S5	2.950	28.92	2.19	16.159	85.31	3.1236×10^{23}	7.3691	5.9385	8.066	639.0

verification that the synthesized samples are indeed glass. The well separated peaks for T_c and T_g in the DTA traces confirms the high thermal stability of the glass samples. The incorporation of increased SrO contents in the glass matrix is evident from the samples color change from light yellow and dark yellow. Consequently, the packing density of the glass becomes higher. The introduction of Sr^{2+} ions into glass allows developing more tightly packed structure by the compaction of electron clouds surrounding the O^- ions due to their charge and coordination number. Both glass compactness and rigidity is affected due to the creation of more non bridging oxygen (NBO).

The FTIR spectra displayed that the incorporation of SrO in lithium borate tends to modify the borate network. It has been observed that SrO content helps in converting more (BO_3) group to (BO_4) units. This reveals that strontium ions have a dominate role in lithium borate glass system. The observed structural changes in the prepared glasses with the increase of Sr^{2+} contents may be due to the enhancement of the degree of localization. As a result, defects are created and driven the energy levels of the closet oxygen ions nearer to the top of the valence band and thereby raise donor center in the glass matrix. The increase in donor center led to a decrease in energy band gap. The enhancement in the number of NBO and large polarizability is responsible for the increase in the refractive index. Non-monotonic change at 7.5% SrO concentration (Figs. 7, 8, 9, 11, and 12) is attributed to the rapid structural change where weak bonds are converted to defects by generating excess NBO. The decrease in optical band gap energy is ascribed to the enhancement of NBO ions in the glass network. The NBO are contributed to valance band maxima via the rupture of metal–oxygen bonds. Consequently, the bond energy is released and the non-bridging orbitals acquire higher energies than the bonding orbitals. The glasses with the smaller Urbach energy possess lower disorder and more compactness.

The physical properties are considerably improved together with the glass compactness due to SrO doping. Addition of SrO is found gradually enhance the glass density. The increase in glass density is attributed to the replacement Li and B by SrO having higher atomic mass. Significant enhancement in the observed field strength is due to the occurrence of strong link between the Sr2+ and B ions. This link attributed to the possible displacement between the Sr2+ and newly generated oxygen atom from the conversion of BO_3–BO_4 units. The intensity of dominant emission is enhanced with the increase of SrO contents. At some SrO concentration luminescence quenching is also evidenced. The quenching in the luminescence intensity beyond 12.5 mol% is primarily attributed to the decrease in SrO distance and the promotion of multi-phonon relaxation. UV–vis-NIR absorption spectra show significant enhancement as a function of increase SrO contents. The samples show wide transparency in the UV range. The increase in gap energy for direct and decrease for indirect transition is attributed to the structural alteration.

5. Conclusion

Strontium ion concentration dependent thermal, structural, physical and optical properties of melt-quench synthesized $Li_4Sr(BO_3)_3$ glasses are reported. Glasses are found to be optically transparent and thermally stable. Their amorphous nature is confirmed by XRD and homogeneity by FESEM measurements. The optical band gap energy, density, molar volume, refractive index, bond vibration frequencies, absorption coefficient, appearance of emission bands, and thermal parameters of these glasses are in good agreement with the previous findings. FTIR spectra revealed the presence of tetrahedral BO_3 and BO_4 groups. Incorporation of SrO is found to affect the glass structure. Meanwhile, the thermal stability is decreased with the increase of SrO contents. The optical absorption edge is observed to shift from 482 to 479 nm. The FWHM for all emission bands is found to increase with the increase of SrO concentration. Physical parameters are strongly influenced by the modifier concentrations. The admirable features of the results suggest that the present glass composition is potential for the photonic devices and optical fibers.

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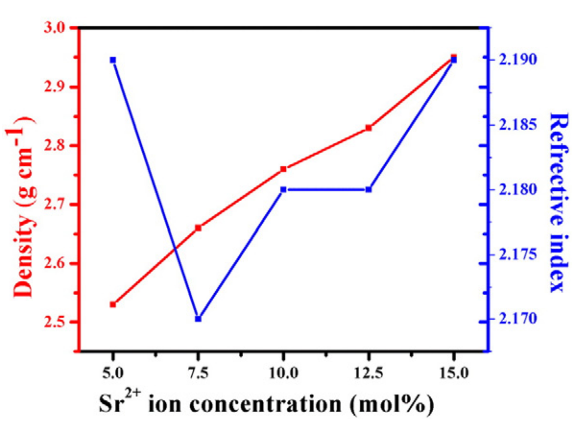

Fig. 12. Density and refractive index variation with SrO concentrations.

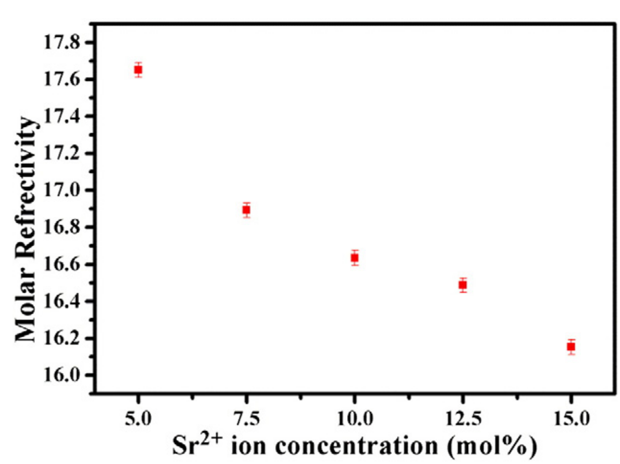

Fig. 13. SrO concentrations dependent variation of molar refractivity.

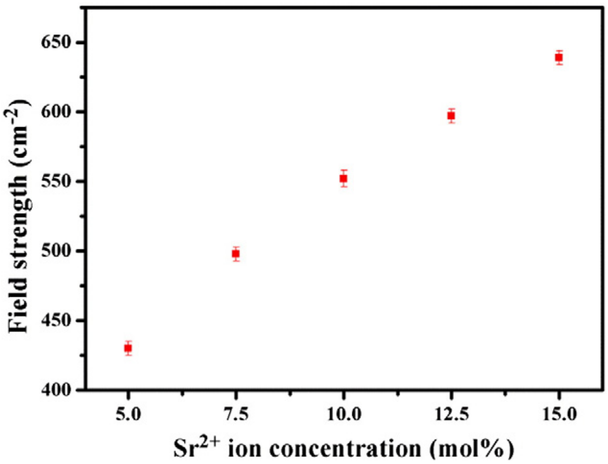

Fig. 14. SrO concentrations dependent field strength alterations.

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